University of Toronto – Faculty of Arts and Science
MAT136 - Calculus 2 - Winter 2022
Final Exam - April 18, 2022

Time allotted: 180 minutes.

Aids permitted: Calculator without integral ($\int$) symbol. non-programmable and non-graphical

Total marks: 100

Last Name: ____________________ **First Name:** ____________________

Student Number: ____________________ **UTORid:** ____________________

Email: ____ @mail.utoronto.ca

Instructions:

- This test contains 16 pages and a detached booklet with the multiple choice questions and a formula sheet.

DO NOT DETACH ANY PAGES.

- Please write your Student Number (not your name) at the top of every page in this booklet.
- Only non-programmable and non-graphical calculators are allowed.
- Cellphones, smartwatches, or any other electronic devices are not allowed. They must be turned off and in your bag under your desk or chair. These devices may *not* be left in your pockets.

Test Writing Instructions:

- **Write clearly** and concisely in a linear fashion. Explain all your reasoning. Do not submit messy scrapwork.
- **Unsupported answers to Questions 14–17** will not receive full credit. A correct answer without explanation will receive no credit unless otherwise noted; an incorrect answer supported by substantially correct calculations and explanations may receive partial credit.
  - Include units in your answers where appropriate.
  - Organize your work in a reasonably neat and coherent way.
  - You must use the methods learned in this course to solve all of the problems.
- For questions with a boxed area, ensure your answer is completely inside the box.
- DO NOT START the test until instructed to do so.

GOOD LUCK!

MULTIPLE CHOICE PART. There is only one correct answer for each question. **(40 marks)**

ANSWER THESE QUESTIONS ON **PAGE 3** OF THE TEST.

1 (4 marks) A function $f(x)$ is continuous and satisfies $\int_{-1}^{2} f(x)\,dx = -3$.

Which of the following could describe an antiderivative $F(x)$ of $f(x)$?

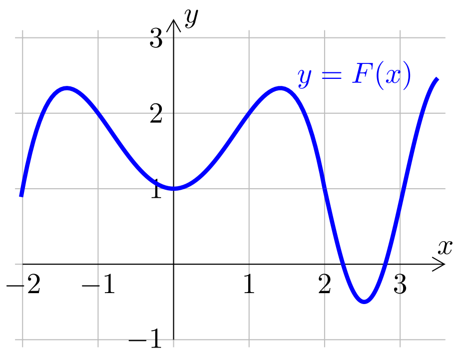
(A)

x	-2	-1	0	1	2	3
$F(x)$	15	12	9	6	3	0

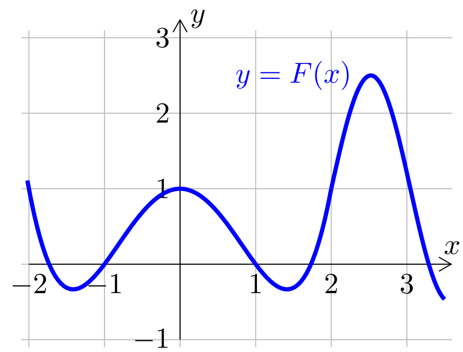
(C)

x	-2	-1	0	1	2	3
$F(x)$	3	4	5	-5	1	0

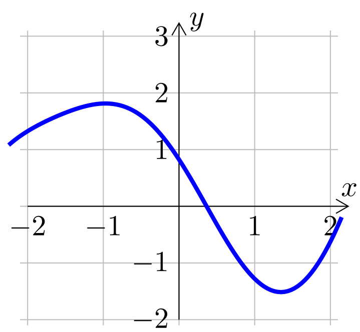
(B)



(D)



2 (4 marks) Consider the Taylor polynomial of $f(x)$ centred around $x = 1$ of degree 3 $p(x) = a + b(x-1) + c(x-1)^2 + d(x-1)^3$, where $f(x)$ is sketched in the figure below:



Which of the options must be true?

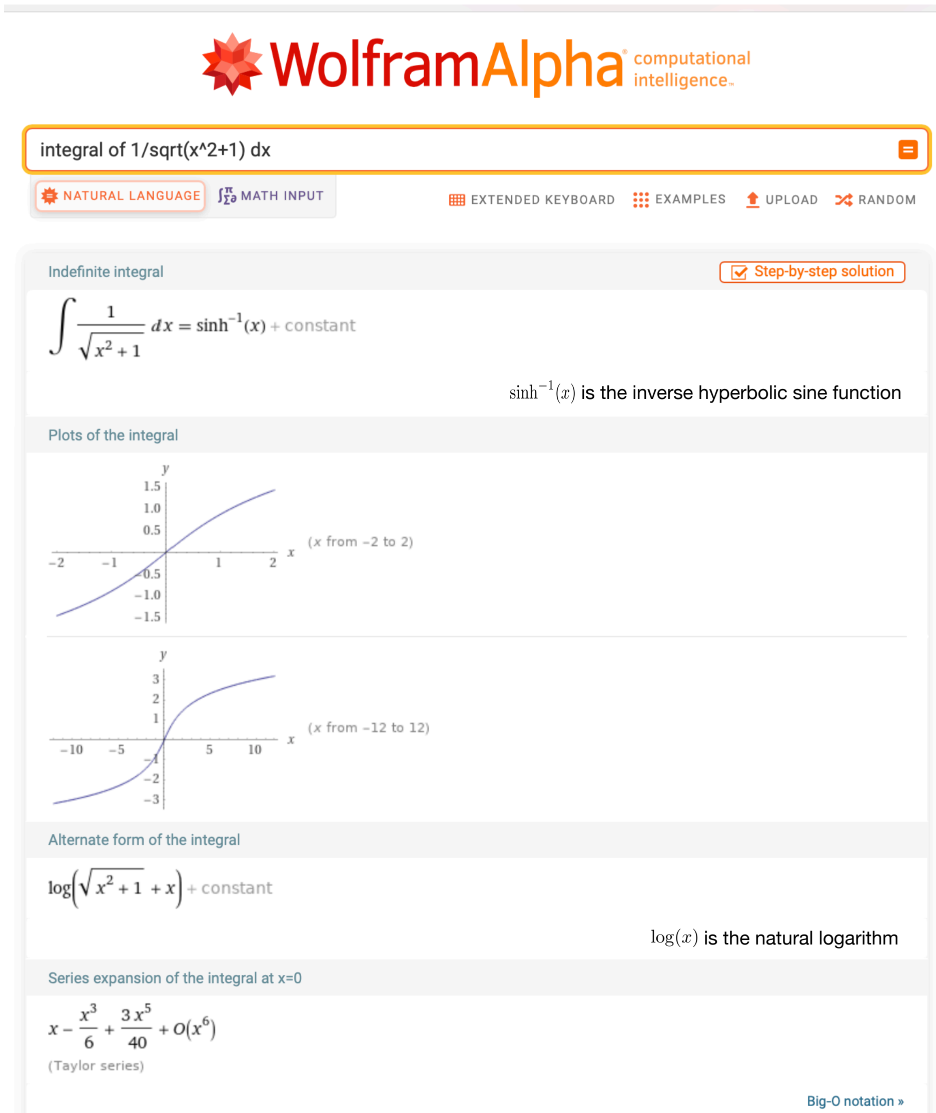
(A) $c < 0$

(B) $c = 0$

(C) $c > 0$

(D) We can't determine the sign of c from this information.

3 (4 marks) Craig is writing an open book online exam for MAT136 and he wants to find the **exact value** of the integral $\int_0^1 \frac{1}{\sqrt{x^2+1}}\, dx$. He goes to WolframAlpha:



How should Craig make sense of this result?

(A) He should write the solution: $\sinh^{-1}(x)$.

(B) He should write the solution: $\sinh^{-1}(0) - \sinh^{-1}(1)$.

(C) He should write the solution: $\ln\left(\sqrt{2}+1\right)$.

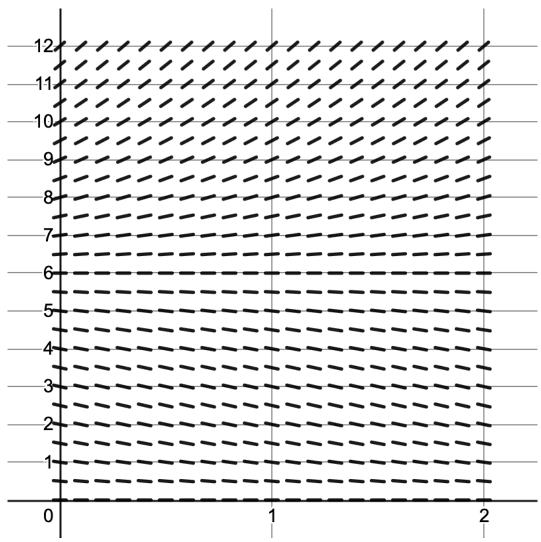
(D) He should write the solution: $\ln\left(\sqrt{x^2+1}+x\right)$.

(E) He should write the solution: Approximately equal to $1 - \frac{1}{6} + \frac{3}{40}$, using the Taylor series provided.

4 (4 marks) According to the Allee effect, a goldfish can have a higher chance of survival in a tank, if there are more goldfish around it. The population of N goldfish is modelled by

$$\frac{dN}{dt} = N\left(\frac{N}{6} - 1\right)\left(1 - \frac{N}{20}\right)$$

and initially there are $N(0) = 7$ gold fish.



Judging by the slope field above, an approximation of the number of goldfish in the tank at $t = 2$ by Euler's method with step $\Delta t = 0.5$ would be...

(A) an under-estimate, because the solution seems to be concave up

(B) an under-estimate, because the solution seems to be concave down

(C) an over-estimate, because the solution seems to be concave up

(D) an over-estimate, because the solution seems to be concave down

5 (4 marks) Which differential equation best describes the following scenario?

A gardener is filling a pot with water at a constant rate. At the same time, the pot is leaking at a rate proportional to the amount of water in the pot.

Let $W(t)$ be the amount of water in the pot, t time, and a, b **positive** constants.

(A) $\dfrac{dW}{dt} = a + bW$

(B) $\dfrac{dW}{dt} = a - bW$

(C) $\dfrac{dW}{dt} = -a + bW$

(D) $\dfrac{dW}{dt} = -a - bW$

6 (4 marks) The amount of hairballs in a cat's belly at time $0 \leqslant t \leqslant 2.5$ is given by

$$f(t) = \int_{(t-1)^2}^{\pi} \big(3s - \sin(3s)\big)\, ds.$$

At what time will there be a maximum amount of hairballs in the cat's belly?

Hint. Recall that for $z > 0$, we have $\sin(z) < z$.

(A) $t = 0$

(B) $t = 1$

(C) $t = \frac{\pi}{2}$

(D) $t = 2.5$

7 (4 marks) Consider a series $\sum_{n=0}^{\infty} c_n (x-2)^n$, which converges when $x = 3$, but diverges when $x = -3$. Which of the following statements **MUST** be true?

(A) It converges when $x = 0$.

(B) It converges when $x = 1$.

(C) It converges when $x = -2$.

(D) It converges when $x = -4$.

(E) None of the above.

8 (4 marks) Consider the following function:

$$f(x) = \int_0^x \frac{\sin(t)}{t}\, dt\,.$$

Which of the following options is the Taylor series of $f(x)$ centred at 0?

(A) $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n\, x^{2n+1}}{(2n+1)!\,(2n+1)}$

(B) $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n\, x^{n+1}}{(2n+1)!\,(n+1)}$

(C) $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n\, x^{2n}}{(2n+1)!}$

(D) $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n\, t^{2n}}{(2n+1)!}$

9 **(4 marks)** The populations of lions and antelopes in a savannah are governed by the differential equations

$$x'(t) = 5x - xy \qquad \text{and} \qquad y'(t) = -4y + 2xy$$

Given that lions eat the antelopes (and not the other way around), which quantity describes the **number of lions** over time?

(A) $x(t)$

(B) $y(t)$

10 **(4 marks)** A lion spots an antelope at a distance and starts to chase it:

- t is the time in seconds since the beginning of the chase;
- $v_L(t)$ is the velocity of the lion in m/s;
- $v_A(t)$ is the velocity of the antelope in m/s;
- $d_L(t)$ is the **distance travelled** by the lion in m;
- $d_A(t)$ is the **distance travelled** by the antelope in m.

What is the **total displacement** of the lion when the antelope has travelled x metres?

(A) x metres.

(B) $\displaystyle\int_0^x v_L(t)\,dt$ metres.

(C) $d_L\big(d_A^{-1}(x)\big)$ metres.

(D) $\displaystyle\int_0^{d_A^{-1}(x)} v_L(t)\,dt$ metres.

Student Number:

MULTIPLE CHOICE PART

Indicate your answers for questions **1–10** on the multiple choice question sheet by completely shading the bubble corresponding to the answer.

Good: ● Bad: ◐ ⊙ ⊗ ✓

1 Ⓐ Ⓑ Ⓒ Ⓓ

2 Ⓐ Ⓑ Ⓒ Ⓓ

3 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

4 Ⓐ Ⓑ Ⓒ Ⓓ

5 Ⓐ Ⓑ Ⓒ Ⓓ

6 Ⓐ Ⓑ Ⓒ Ⓓ

7 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

8 Ⓐ Ⓑ Ⓒ Ⓓ

9 Ⓐ Ⓑ

10 Ⓐ Ⓑ Ⓒ Ⓓ

GRAPHING PART – You don't need to justify these questions and you don't need to have a formula for the functions you sketch. Make sure your sketches are clearly visible.

11 (3 marks) Consider the differential equation

$$\frac{dy}{dx} = (y+1)(1-y)^2.$$

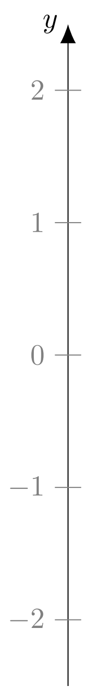
For each of the following conditions, sketch the graph of one solution that satisfy it and label it clearly (there should be three graphs in your solution) (a formula is not required)

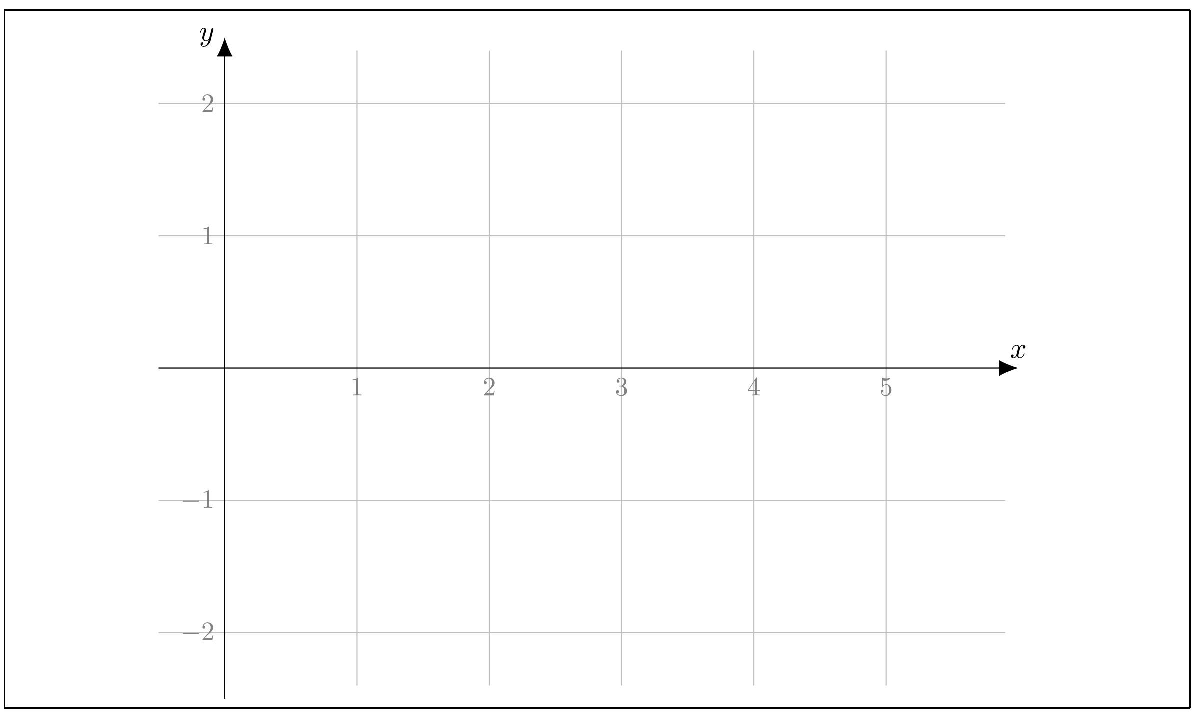
(a) $y(0) = -1$

(b) Passes through the point $(1, \frac{3}{2})$

(c) $\lim_{x\to\infty} y(x) = -\infty$

Hint. You can use the vertical y axis on the left to help you with the sketch, but it won't be graded.

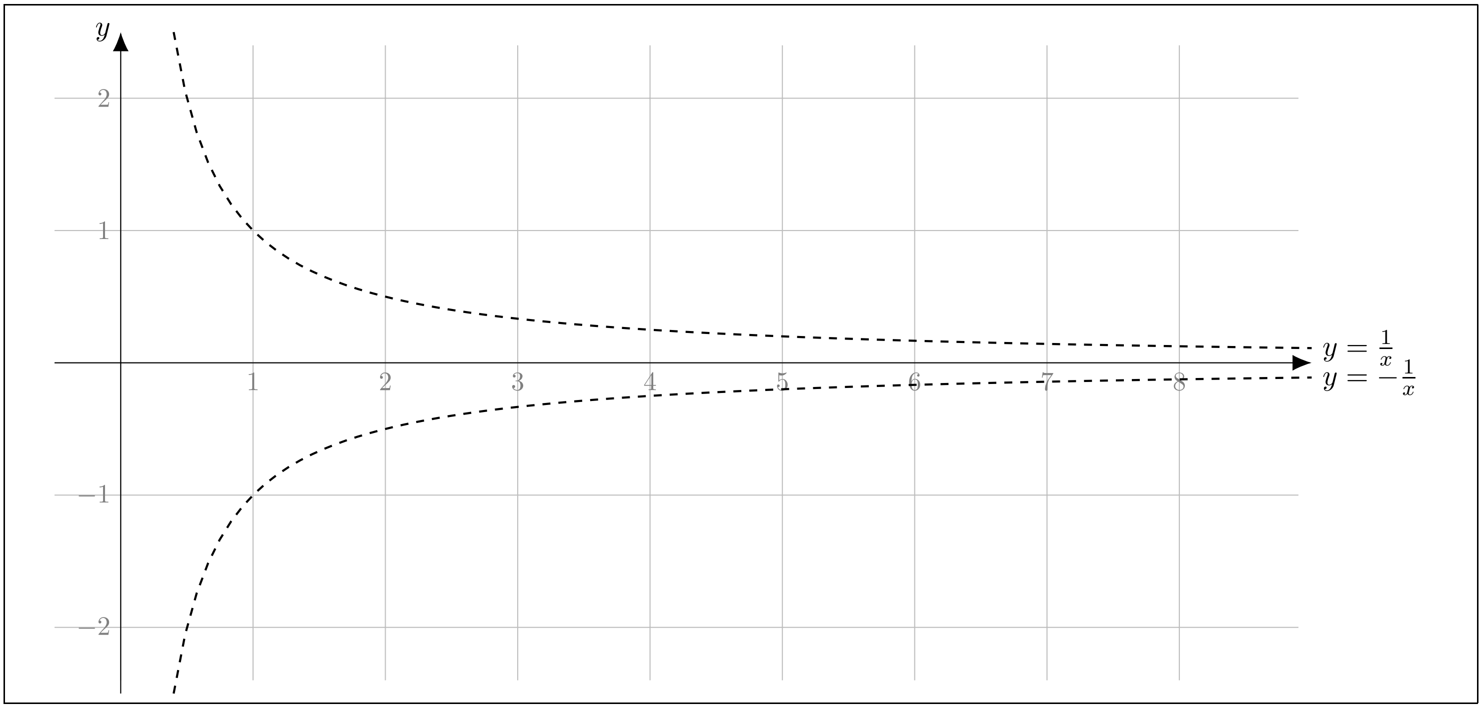




12 (5 marks) Sketch the graph of a function $f(x)$ that satisfies the following properties:

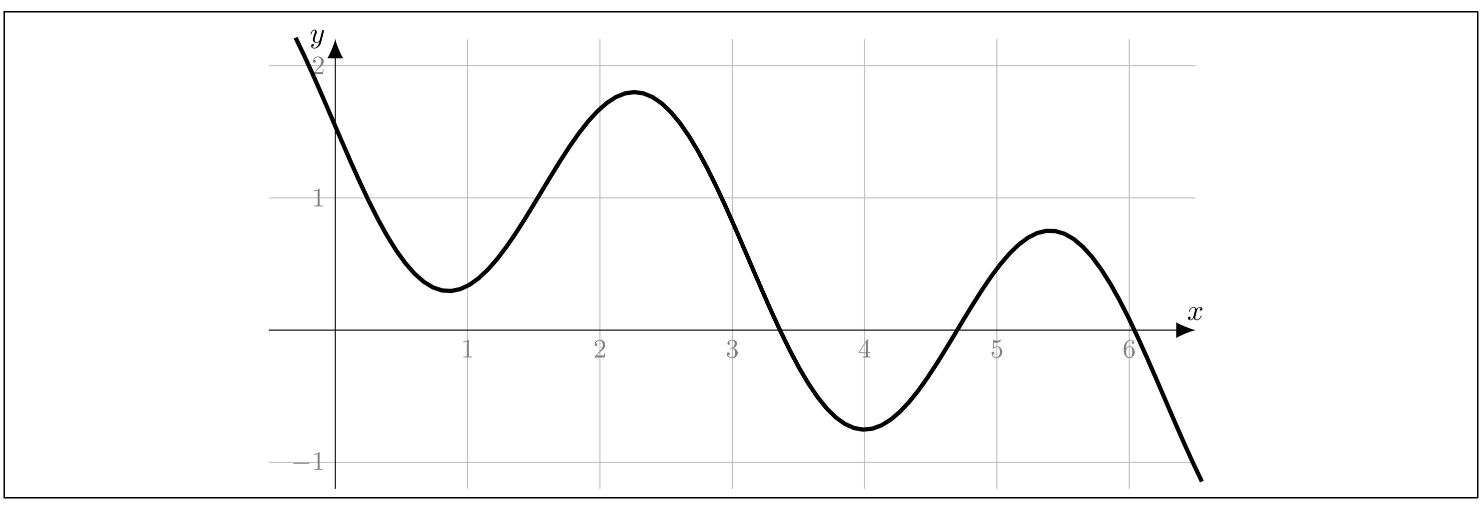
(a formula is not required)

- $f(x)$ is continuous for all $x \geqslant 0$
- $f(2)$ and $f(4)$ have opposite signs (and are not zero)
- $\lim_{x\to+\infty} f(x) = 0$
- $\int_1^{+\infty} f(x)\, dx = \infty$



13 (3 marks) Consider the graph below of a function $f(x)$.

Sketch its Taylor polynomial of degree 2 centred at $x = 1$. (a formula is not required)



LONG ANSWER PART – Unsupported answers to Questions **14–17** will not receive full credit.

14 Bispectral index (BIS) is measure of the level of consciousness through a patient's **(11 marks)** electroencephalogram (brain activity). It is commonly used in surgeries as one of the tools to monitor the patient's depth of anesthesia:

- BIS = 100 means that the patient is fully awake
- $40 \leqslant \text{BIS} \leqslant 60$ is an appropriate level for general anesthesia

In a scientific paper[1], the authors collected data on how the amount of Propofol (a common anesthetic) relates to BIS.

Consider the functions:

- $\text{BIS}(t)$ = the BIS level of the patient t minutes into the surgery
- $\frac{dP}{dt}$ = the rate that Propofol is being administered to the patient, in μg/minute.

The data collected during the first 25 minutes of one surgery is transcribed below.

t (in minutes)	0	1	2	3	$\cdots$	7	8	9	10	$\cdots$	25
$\frac{dP}{dt}$ (in μg/min)	0	5	15	25	25	25	25	$\cdots$	25	$\cdots$	25
BIS	97	90	78	80	$\cdots$	80	72	64	57	$\cdots$	55

Note that human beings do not produce Propofol, so before the surgery the patient's body doesn't have any Propofol.

14.1 (3 marks) Explain the meaning of $P(t)$ in practical terms.

[1] "Comparative Evaluation of a New Depth of Anesthesia Index in ConView® System and the Bispectral Index during Total Intravenous Anesthesia: A Multicenter Clinical Trial", by Fu, et al, `https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6425335/`

14.2 **(4 marks)** Dr. Paul, the anesthesiologist needs to calculate the total amount of Propofol used during the surgery. Estimate from the table above the amount of Propofol that was used during the first 25 minutes of surgery.

Total amount of Propofol $\approx$

14.3 **(4 marks)** After studying the data, Dr. Paul found a formula that gives an approximation for the function $\mathrm{BIS}'(P)$:

$$\frac{d\,\mathrm{BIS}}{dP} \approx -\frac{45}{2}\sqrt{\frac{10}{(P+10)^3}}$$

Assuming this formula is exact and using $\mathrm{BIS}(0)$ from the table, find the formula for $\mathrm{BIS}(P)$.

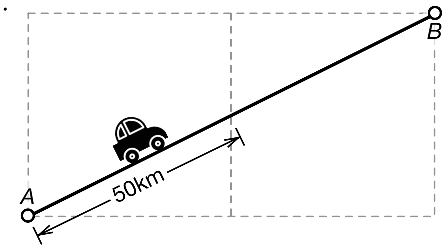
$\mathrm{BIS}(P) =$

15 Kim and Murray are on a road trip with their car. The car's fuel consumption is **(13 marks)**

given by $C(s) = 10 + s$ **litres per 100km**. For example, on a flat road, the consumption is $C(0) = 10$ and on a road with downhill slope $s = -\frac{1}{2}$, the consumption is $C(-\frac{1}{2}) = 9.5$.

15.1 **(3 marks)** Consider the stretch of road as depicted on the right. How much fuel will be consumed on the drive from A to B?

You do NOT need to justify your answer to this part.

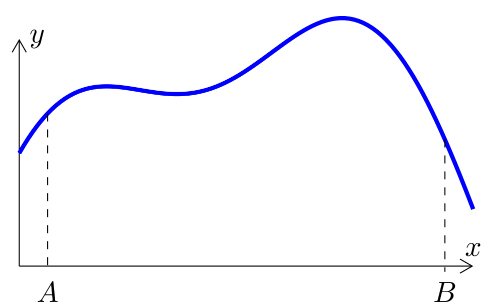


Fuel consumed =	litres

15.2 **(10 marks)** Now consider a mountain range whose shape is given by a differentiable function $y = f(x)$. Both x and y are given in km. An example of such a function is given on the right.

Find an integral formula for the car's total fuel consumption in litres driving from A to B.

In your answer, follow the step-by-step slicing procedure (in the formula sheet).



Note. Leave your answer in integral form. Do not compute the integral.

Fuel consumed = litres

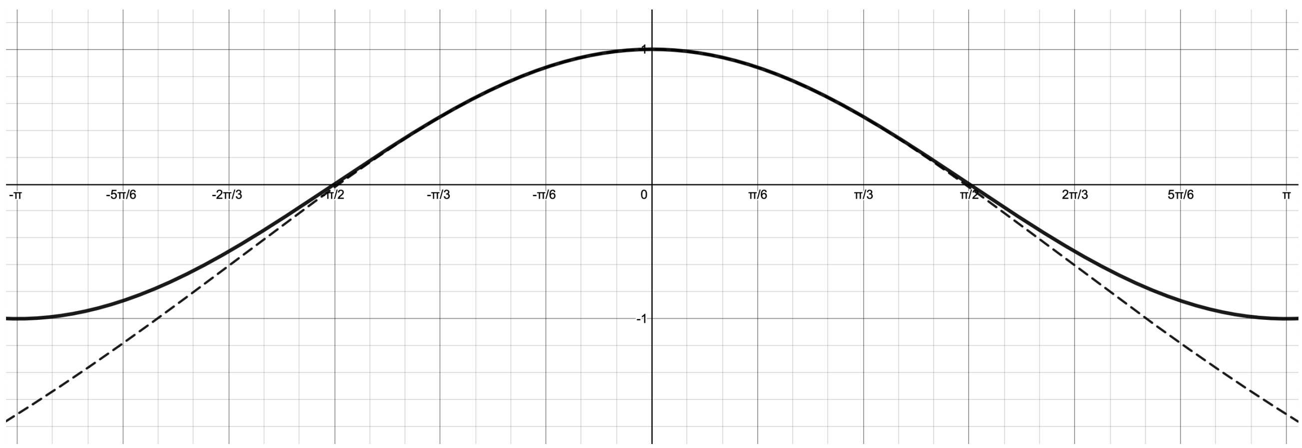
15.3 **(2 bonus marks)** Choose an appropriate function $f(x)$ in **15.2** and verify that your result in **15.1** is a special case of your result in **15.2**.

16 In this function we are going to find an approximation for $\cos(x)$ using a new approach that you have not seen before. **(12 marks)**

We want to want to approximate $f(x) = \cos(x)$ near $x = 0$ by a simple function of the form

$$g(x) = \frac{1 + a \cdot x^2}{1 + b \cdot x^2},$$

as in the figure below, where the dashed line is the approximation above and the solid line is $\cos(x)$.



The goal of this question is to find the best parameters a and b.

16.1 **(4 marks)** Find the Taylor series for $\frac{1}{1+bx^2}$ centred at $x = 0$. In your answer, use Σ notation or provide at least three nonzero terms.

Taylor series for $\dfrac{1}{1+bx^2}$ centred at $x = 0$ is

16.2 **(4 marks)** Use the result from **16.1** to find the **first three nonzero terms** of the Taylor series of $g(x) = (1 + ax^2) \cdot \dfrac{1}{1 + bx^2}$ centred at $x = 0$.

First three nonzero terms:

16.3 **(4 marks)** Find the parameters a and b that guarantee that the first three nonzero terms of the Taylor series you found in **16.2** match the first three nonzero terms of the Taylor series for $f(x) = \cos(x)$ centred at $x = 0$.

$a =$

$b =$

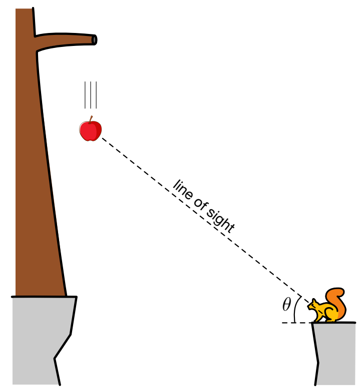
17 A squirrel stares longingly at a falling apple, falling from a tree beyond a deep chasm[2] **(13 marks)**

which is 4m wide.

The squirrel already passed the MAT135 exam, so it knows that the angle θ between the ground and its line of sight satisfies

$$\tan\theta = \frac{y}{4},$$

where y is the current height of the apple, and we assume that $-\frac{\pi}{2} \leqslant \theta \leqslant \frac{\pi}{2}$. Since the apple is falling, this means that

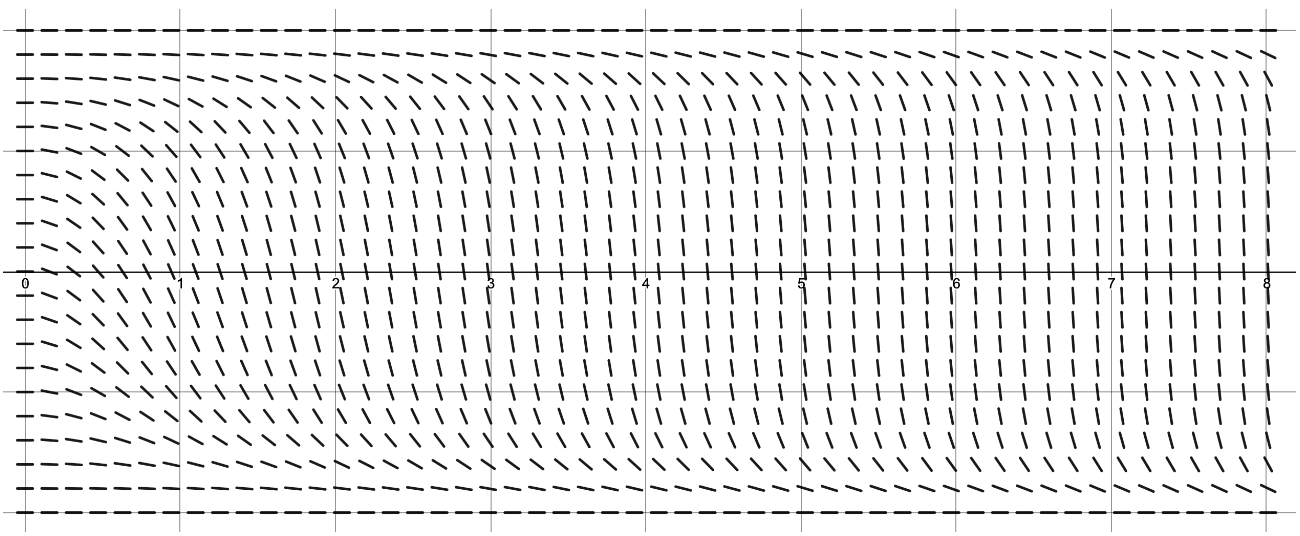


$$\frac{1}{\cos^2(\theta)}\frac{d\theta}{dt} = \frac{1}{4}\frac{dy}{dt},$$

so the angle between the squirrel's line of sight and the apple is changing according to the differential equation

$$\frac{d\theta}{dt} = \cos^2(\theta)\big(-\tfrac{g}{4}t\big),$$

where g is the gravitational acceleration. Below is the slope field for this differential equation.



17.1 **(3 marks)** Recall that equilibrium solutions are constant solutions for all t. Using the differential equation, find **ALL** equilibrium solutions and label them on the vertical axis in the slope field.

Equilibrium Solutions:

[2] a chasm is a deep or seemingly bottomless opening in the earth's surface.

17.2 **(2 marks)** The values $\theta = -\frac{\pi}{2}$ and $\theta = \frac{\pi}{2}$ are not possible in the real world context of the squirrel and the apple. Explain why.

17.3 **(2 marks)** For some real world initial conditions, the solution approaches a negative asymptote as $t \to \infty$. What are those initial conditions? Explain why.

$\boxed{} < \theta(0) < \boxed{}$

17.4 **(3 marks)** Describe the solutions of the differential equation that satisfy the initial condition described in **17.3** in practical terms. What happens to the squirrel and apple for those solutions and how are these related to the values of $\theta(t)$?

17.5 (3 marks) Using separation of variables, find the solution of this differential equation that passes through the point $\left(0, \frac{\pi}{4}\right)$.

Hint 1: The formula sheet might have some useful formulas.

Hint 2: $\tan(\frac{\pi}{4}) = 1$.

$\theta(t) =$

USE THIS PAGE TO CONTINUE OTHER QUESTIONS.

If you wish to have this page marked, make sure to refer to it in your original solution.

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